

HOME WORK 1

Due date: October 9, 2023.

Note: Remember to include the self-assessment grades on the first sheet of your home work (see handout for details).

Reading assignment:

- Read chapter 2 of the class notes (on class web-site).
- Read chapters 1, 2 & 3 of the text book.

Problem assignment:

1. Find a way to represent complex numbers as 2×2 **real** matrices, such that arithmetic operations (+, −, ×, and ÷) on complex numbers become equivalent to arithmetic operations on their matrix representations instead. That is, if $T(z)$ is the 2×2 real matrix representing the complex number z , then

$$\begin{aligned} T(z_1 + z_2) &= T(z_1) + T(z_2), \\ T(z_1 - z_2) &= T(z_1) - T(z_2), \\ T(z_1 z_2) &= T(z_1) T(z_2), \\ T(z_1 / z_2) &= T(z_1) (T(z_2))^{-1}, \\ T(\bar{z}_1) &= (T(z_1))^T, \end{aligned}$$

for all complex numbers z_1 and z_2 . *Hint:* The entries of $T(z)$ will depend on the real and imaginary parts of z . Look at the expression for the product of two complex numbers and two 2×2 matrices.

2. Show that any $m \times n$ complex matrix A can be represented as a $2m \times 2n$ real matrix $S(A)$, such that all the usual matrix operations are preserved:

$$\begin{aligned} S(A + B) &= S(A) + S(B), \\ S(A - B) &= S(A) - S(B), \\ S(AB) &= S(A) S(B), \\ S(A^H) &= (S(A))^T, \\ S(A^{-1}) &= (S(A))^{-1}. \end{aligned}$$

The proof of the last equation is a bit subtle. What happens if you only know that $S(A)$ is invertible?

3.

- A) Show that the exchange operator $E_{i,j}$, which exchanges the i -th and j -th components of a vector leaving all others untouched:

$$E_{i,j}(x) = E_{i,j} \begin{pmatrix} \vdots \\ x_i \\ \vdots \\ x_j \\ \vdots \end{pmatrix} = \begin{pmatrix} \vdots \\ x_j \\ \vdots \\ x_i \\ \vdots \end{pmatrix},$$

is a linear operator.

- B) Find a matrix representation for $E_{i,j}$.

C) Show that the **permutation** operator

$$P_{\sigma_1 \sigma_2 \dots \sigma_n} x = \begin{pmatrix} x_{\sigma_1} \\ x_{\sigma_2} \\ \vdots \\ x_{\sigma_n} \end{pmatrix}, \quad 1 \leq \sigma_i \leq n, \quad \sigma_i \neq \sigma_j \text{ if } i \neq j,$$

is a linear operator and find its matrix representation.

D) How many distinct exchange matrices are there of size $n \times n$?

E) How many distinct permutation matrices are there of size $n \times n$?

F) Show that every $n \times n$ permutation matrix can be written as the product of at most $n - 1$ exchange matrices.

4. Let Π_n denote the set of all polynomials of degree at most n with real coefficients.

A) Find a way to represent any polynomial in Π_n as a real column matrix with $n + 1$ elements. That is, find an invertible map T from Π_n to $\mathbb{R}^{n+1}$ such that $T(p_1 + p_2) = T(p_1) + T(p_2)$ and $T(\alpha p_1) = \alpha T(p_1)$, for any two polynomials p_1 and p_2 from Π_n and any scalar α .

B) Can you find a representation that is different from the one in the previous part? (There are infinitely many).

C) Find an $n \times (n + 1)$ matrix D such that $DT(p) = T(p')$ for any polynomial $p \in \Pi_n$, where p' denotes the derivative of p .

D) Is the matrix D invertible? Left-invertible? Right-invertible?

5. **Quaternions** are complex 2×2 matrices of the form

$$\begin{pmatrix} z_1 & z_2 \\ -\bar{z}_2 & \bar{z}_1 \end{pmatrix},$$

where z_1 and z_2 are complex numbers, and $\bar{z}$ denotes the complex conjugate of z . [Quaternions can be useful (Google is your friend).]

A) Show that if q_1 and q_2 are quaternions, then $q_1 + q_2$, $q_1 - q_2$ and $q_1 q_2$ are all also quaternions.

B) Show that if $q \neq 0$ is a quaternion, then q^{-1} exists and is a quaternion.

C) Find two quaternions q_1 and q_2 such that $q_1 q_2 \neq q_2 q_1$.

D) Define

$$\mathbf{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{i} = \begin{pmatrix} \iota & 0 \\ 0 & -\iota \end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix} 0 & \iota \\ \iota & 0 \end{pmatrix}, \quad \mathbf{k} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

where $\iota^2 = -1$.

i. Show that every quaternion q can be written as $q = a\mathbf{1} + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$, where a, b, c, d are all real numbers.

ii. Give simple expressions for $\mathbf{i}^2, \mathbf{j}^2, \mathbf{k}^2, \mathbf{ij}, \mathbf{ik}, \mathbf{ji}, \mathbf{jk}, \mathbf{ki}, \mathbf{kj}$.

E) Show how to represent the quaternions as 4×4 real matrices such that the algebraic operations are preserved.